ADVANCED PROGRAMMING

Laboratory Practice (CATL)

Ordinary Call – May 2026

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The Battle of Hawkins: Simulation of the invasion of the Upside-Down

Just another day at the office

Part 1: Concurrent Programming

In the fall of 1987, the city of Hawkins (Indiana) is under threat from Vecna and the creatures that inhabit the parallel dimension of the Upside Down. The children of Hawkins must organize themselves to withstand the attacks of the demogorgons, while Eleven and her companions struggle to keep the portals under control. The task is to design and implement a system that allows managing and coordinating all the elements involved. The general characteristics of the system are as follows:

- In the system there are two types of users: children and demogorgons.
- There are two interconnected dimensions: the normal world of Hawkins, which acts as a safe zone, and the Upside Down, the dimension where the demogorgons roam.
- The Upside Down has 4 distinct unsafe zones (FOREST, LABORATORY, SHOPPING MALL, AND SEWER). In addition, there is a zone called HIVE that will serve as a storage for victims.
- The children must be created (instantiated) in a staggered manner (that is, they must be created gradually) at random intervals between 0.5 and 2 seconds. The children will begin their life on Hawkins' Main Street.
- There are 4 PORTALS that connect Hawkins with each of the unsafe zones. Only one child can pass through each portal at a time. If there are children on both sides of the portal, priority will always be given to the one trying to return to Hawkins.
- The city of Hawkins has 3 distinct zones: MAIN STREET, BYERS' BASEMENT, and WSQK RADIO.
- The system will generate 1,500 children and 1 initial demogorgon (the Alpha Demogorgon).

DEMOGORGONS

All the demogorgons in the system must necessarily be modeled as threads and will be identified as "DXXXX", where X is a unique number (id), such as D0001, D0023, D1234, etc. The only demogorgon initially generated (the Alpha Demogorgon) will have the id "D0000".

The demogorgons will move between the different unsafe areas of the Upside Down in search of children, moving randomly from one to another. Upon arriving at an area, they will check if there is any child present: if there is at least one, they will randomly select a target and attempt to capture it, a process that will last a random duration between 0.5 and 1.5 seconds. If the attack is successful, the child will be taken to the HIVE; once inside, it will take a random time between 0.5 and 1 second to deposit the child, and the demogorgon's capture counter will increase. If, on the other hand, the attack fails, the child will remain in the area. In either case, after the attack, the demogorgon will randomly move to another unsafe area. Conversely, if upon arriving at the area there is no child present, the demogorgon will remain there for a random time between 4 and 5 seconds before randomly moving to another unsafe area and repeating the process.

For every 8 children captured in the HIVE, Vecna will generate a new demogorgon.

This behavior repeats indefinitely.

CHILDREN

Children must be mandatorily modeled as threads and will be identified as "CXXXX", where X is a unique number (id), such as, for example, C0001, C0023, C1234, etc.

For safety, all children will begin their activity on the MAIN STREET of Hawkins. Once there, they will iteratively repeat the following behavior cycle:

- They will access the BYERS' BASEMENT, where they will prepare for exploring the Upside Down. They will remain there for a random time between 1 and 2 seconds.
- From the BYERS' BASEMENT, they will select a PORTAL to exit to the Upside Down and will wait until a group of the size required by that specific portal is formed (2 for the portal leading to the FOREST, 3 for the portal leading to the LABORATORY, 4 for the portal leading to the SHOPPING MALL, and 2 for the portal leading to the SEWER). Once the group is formed, each child will cross the portal one by one (estimated time: 1 second). The group is exclusive: if an additional child arrives while the group is forming or crossing, they must wait to form the next group.
- In the Upside Down, they will remain for a random time between 3 and 5 seconds, and each will collect 1 unit of Vecna's blood. When a child collects the blood and has not been attacked, they will return to the portal to go back to Hawkins without waiting for their companions.
- If a child is attacked by a demogorgon, they will attempt to resist throughout the entire attack. The duration of the attack is determined by the demogorgon, so the child cannot resume movement until the attack ends. The probability of resisting the attack is 2/3.
- If the resistance fails, the child is CAPTURED by the demogorgon and taken to the HIVE.
- The demogorgon that captured the child increases its capture counter. While the child is captured, the demogorgon can continue attacking other children in the same area.
- Once back in Hawkins, they will automatically deposit the collected blood and go to RADIO WSQK, remaining there for a random time between 2 and 4 seconds.
- After resting, they go to MAIN STREET and wander around for 3 to 5 seconds to avoid raising suspicion among the army.
- After completing these activities, they return to the BYERS' BASEMENT to repeat the cycle.

RANDOM GLOBAL EVENTS

The system will periodically generate global events that will affect the behavior of all active threads simultaneously.

Events occur at random intervals between 30 and 60 seconds from the start of the system and from the end of the previous event. That is, every 30–60 seconds a new event will occur.

Each event has a random duration of between 5 and 10 seconds. The possible types of events are:

- LABORATORY BLACKOUT:

o All portals become temporarily unusable (no child will be able to cross them). Children already in transit through a portal finish crossing normally. Those waiting to cross must wait until the blackout ends.

o Demogorgons will continue attacking children, but they cannot change zones during the blackout and remain in the zone where they are located.

- UPSIDE DOWN STORM:

o The blood collection time of all children in the Upside Down is doubled during the event.

o Demogorgons become more aggressive: the time between attacks is reduced by half.

- ELEVEN'S INTERVENTION:

o Eleven uses her powers to free as many children captured in the HIVE as units of Vecna's blood have been collected during that time, and they return to the MAIN STREET of Hawkins.

o During this event, all demogorgons are paralyzed (they do not attack or move) for the duration of the event.

- THE HIVE MIND:

o The Mind Flayer connects psychically with all active demogorgons for the duration of the event. While the connection lasts, all demogorgons move toward the zone where the most children are at that moment.

Other additional considerations:

- All system behavior will be displayed graphically on screen through a Java user interface. The interface will show the real-time status of each zone, the children and demogorgons present, the available blood, and the status of the active global event (if any).
- The operation of all children and demogorgons never ends.
- The children carry a Walkie-Talkie, so they will continuously report on the system's behavior, and this will be saved in a log file called "hawkins.txt". Each line will include a timestamp (date and time with exact second) and the event. Examples: "Demogorgon D0002 attacks child C1234 (captures: 2)", "Child C4567 has been captured", "GLOBAL EVENT: Laboratory blackout started". The logging system will be implemented as a shared resource and will be properly protected.

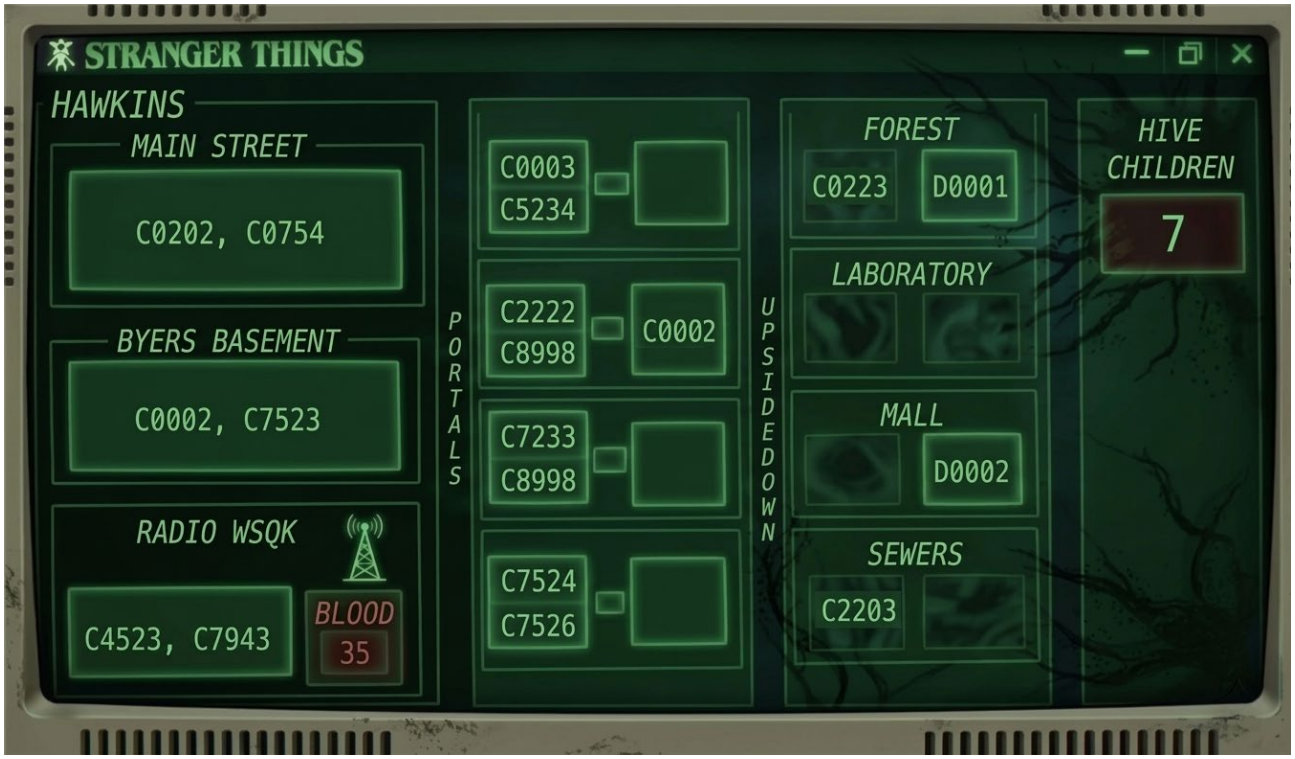

Figure 1. Example of an interface. The design is for illustrative purposes only; what matters is the data displayed, not the visual appearance.

***Note**: The IDs and quantities shown in the interface examples may not be accurate or complete. The images are provided solely to give you an idea of what the final interface will look like and the information it should contain.*

Part 2: Distributed Programming

Based on the previous Part 1, include a new remote access module implemented using RMI or Sockets, which provides the following information and operations:

- Number of children in Hawkins.
- Number of children in each of the portals.
- Number of children in each zone of the Upside Down (including how many are captured in the HIVE).
- Number of demogorgons in each zone of the Upside Down.
- Ranking of the 3 demogorgons with the highest number of captures.
- Status of the global event active at that moment (type of event and estimated remaining time), or "No active event" if there is none.
- Button to stop/resume the execution of the main program.

The retrieval of this data must be in real time and must be displayed automatically, that is, without the user having to press any button or interact in any way to update the information.

One possible example of a graphical interface is shown in the following figure:

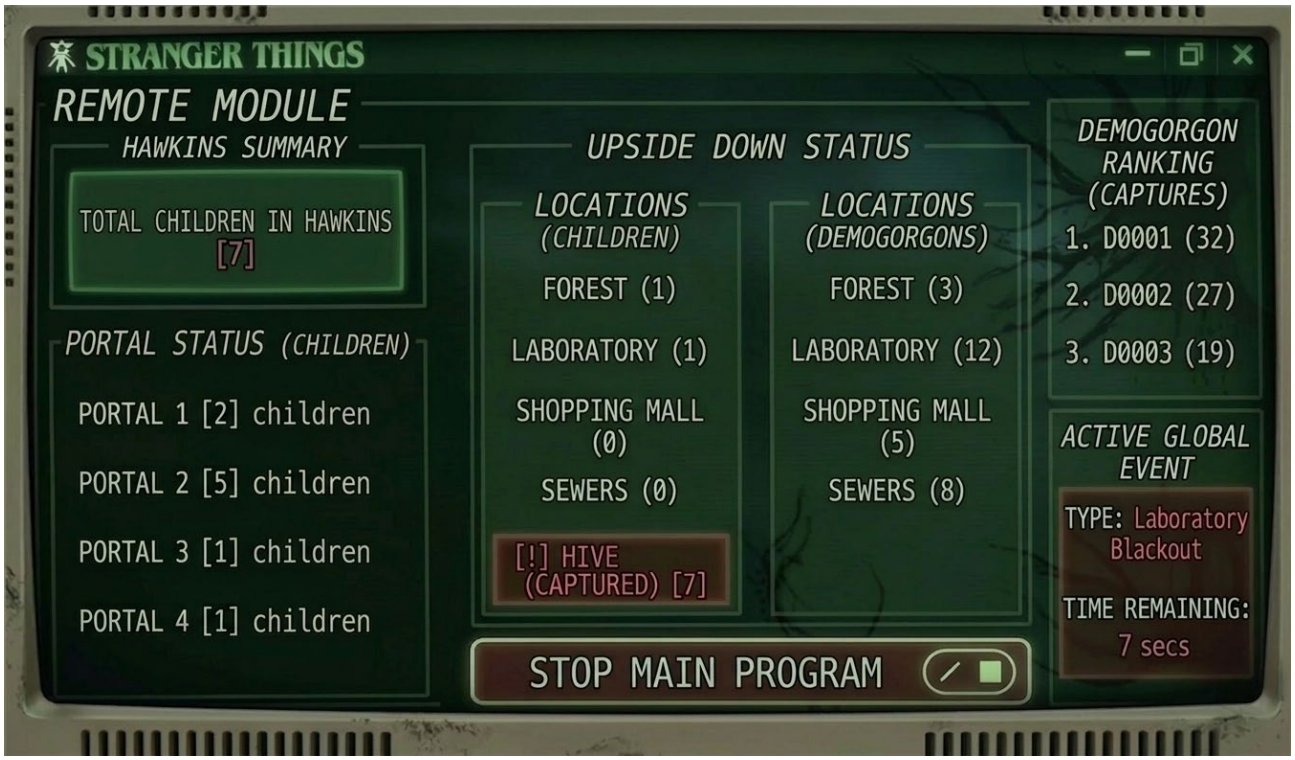

Figure 2. Sample illustration of the remote client interface.

***Note**: The images are shown solely to give an idea of the final interface and the information it should contain, not its visual appearance.*

Delivery conditions

1. *In total, two programs must be developed:*
    a. *A server, whose base code will be the program developed in Part 1, extended with the corresponding functionality to support the distributed programming module.*
    b. *A client program that allows performing the operations detailed in Part 2, to check the system status automatically and remotely pausing/resuming execution.*
2. *All the mechanisms used in class can be used to solve all the communication and synchronization problems that arise in this statement. However, the synchronization and communication mechanisms that solve the problem in the* ***most efficient and optimal way*** *possible must be used.*
3. *The practice will be mainly done* ***in pairs*** *(it can also be done individually) and must be delivered* ***before the date*** *indicated in the Virtual Classroom, through the corresponding task, by uploading two files: the* ***practice report*** *in PDF format and the complete* ***NetBeans project****, compressed as* ***ZIP*** *(do not use the .rar extension).*
4. *The delivery outside the term indicated in the Virtual Classroom will mean a reduction in the final grade, being 25% if it is delivered the next day to the deadline, or 50% if it is delivered within the next two days. Delivery beyond these two days will not be admitted under any circumstances.*
5. *Those practices that does not contain a* ***NetBeans project*** *will not be accepted. The use of any other different IDE will imply non-acceptance of the practice.*
6. *To pass, it is a necessary condition that all programs work correctly and according to the specifications indicated in the statements.*
7. *The solution must be developed using* ***good programming practices****. For example, it is necessary that all the names of the classes begin with a capital letter and all the names of attributes and methods begin with a lowercase letter; the attributes must be private and can only be accessed through getter and setter methods.*
8. ***Both parts*** *(Part 1 and Part 2) of laboratory practice* ***should be delivered together*** *(i.e., in a single project and a single report), as Part 2 is built on Part 1.*
9. *If the exercise is done by a couple,* ***only one of the members must upload it*** *to the virtual classroom, indicating the name of both students in the cover of the report.*
10. *The following data must appear on the cover of the report:*
    a. ***Computer Science Degree/ Computer Engineering Degree/ Double Degree in Computer Science and Business Administration and Management/ Foreign student***
    b. ***Course 2025/2026 – Extraordinary final practice assignment***
    c. ***DNI/Passport – Surname, Name***
11. ***The report must include, as an annex, the source code*** *of the program. If this were not the case, the practice cannot be approved.*
12. *The explanatory report of the practice done must include, in the following order: 1) a high-level analysis (general description of the problem and identification of its main actors); 2) general design of the system and discussion of the synchronization tools used; 3) description of the main classes (attributes and methods); 4) class diagram; and 5) the source code, as an annex.*

13. *Said documentation, except for the code, should not extend more than 20 pages. The **quality of the documentation** - presentation, structure, content, writing - will be a basic element in the evaluation of the practice.*
14. *For the defence/practice exam, the student may take a printed copy of the memory delivered with the practice. It can be the complete memory, or just the source code. For the defence, the use of electronic devices is not permitted (including for consulting the memory). During the defence/exam, the memory cannot be shared between classmates.*
15. *The resolution of the laboratory practice must be genuine and done from scratch, i.e., it is not possible to use any type of source code from other practices submitted to this or previous calls, whether your own or that of other colleagues. If this situation is detected, the practice will not be evaluated and will have a grade of 0. In addition, the corresponding procedures will be executed to communicate such fraudulent behaviour, with academic consequences in accordance with current legislation.*